COUNTER-EXAMPLES TO $\operatorname{Log} C_{3,2}$ IN CHARACTERISTIC 2 AND 3

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ABSTRACT. In this note, we construct a separable surjective morphism $f : X \to S$ with connected fibers over an algebraically closed field of characteristic $p = 2$ or 3 from a smooth projective 3-fold X to a smooth projective surface S and a smooth prime divisor B on X such that

$$\underbrace{\kappa\left(X, K_X + \frac{2}{p}B\right)}_{-\infty} < \underbrace{\kappa\left(X_\eta, K_{X_\eta} + \frac{2}{p}B\Big|_{X_\eta}\right)}_{0} + \underbrace{\kappa(S)}_{0 \text{ or } 1},$$

where X_η is the generic fiber of f.

1. Introduction

The following is a well-known conjecture proposed by Iitaka [13].

Conjecture 1.1. *Let k be an algebraically closed field of characteristic 0. Let $f : X \to Y$ be an algebraic fiber space between smooth projective varieties with $\dim X = n$ and $\dim Y = m$. Then*

$$C_{n,m} : \kappa(X) \geq \kappa(X_\eta) + \kappa(Y),$$

where X_η is the generic fiber of f.

Here, by an algebraic fiber space we mean a separable surjective morphism with connected fibers. This conjecture has been studied by numerous mathematicians for half a century. For instance, see [21, 22, 23, 14, 24, 15, 16, 9, 2, 17, 10, 11, 12].

Conjecture 1.1 has also been studied in positive characteristic [4, 19, 5, 8, 1, 25, 3]. Recently, Cascini–Ejiri–Kollár–Zhang [3] constructed algebraic fiber spaces of positive characteristic that violate $C_{n,m}$. The smallest n in their work is 4 (when $p = 2$). In this note, we construct algebraic fiber spaces *with boundary divisors* that violate $\operatorname{Log} C_{3,2}$, the following natural logarithmic generalization of $C_{3,2}$.

Problem 1.2. *Let k be an algebraically closed field. Let (X, Δ) be a projective lc pair of dimension n and let $f : X \to Y$ be an algebraic fiber space to a smooth projective variety Y of dimension m. Does the following inequality hold?*

$$\operatorname{Log} C_{n,m} : \kappa(X, K_X + \Delta) \geq \kappa(X_\eta, K_{X_\eta} + \Delta|_{X_\eta}) + \kappa(Y).$$

Theorem 1.3. *There exists an algebraic fiber space $f : X \to S$ over an algebraically closed field of characteristic $p = 2$ or 3 from a smooth projective 3-fold X to a smooth*

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projective surface S and a smooth prime divisor B on X such that

$$\underbrace{\kappa\left(X, K_X+\frac{2}{p}B\right)}_{-\infty} < \underbrace{\kappa\left(X_\eta, K_{X_\eta}+\frac{2}{p}B\Big|_{X_\eta}\right)}_{0} + \underbrace{\kappa(S)}_{0 \text{ or } 1}.$$

Here, X_η is the generic fiber of f.

It is worth mentioning that if $p=3$, then $(X, \frac{2}{p}B)$ is klt.

Remark 1.4. The reason why there exist counter-examples to $\operatorname{Log} C_{n,m}$ may be a lack of a suitable assumption. For example, under the assumption that pairs induced on general fibers have only mild singularities, we have not found counter-examples to $\operatorname{Log} C_{n,m}$. Indeed, general fibers of algebraic fiber spaces in [3] have a cusp, and in Theorem 1.3, general fibers are isomorphic to $\mathbb{P}^1$ but the coefficient of the boundary restricted to general fibers is 2.

Remark 1.5. There are several extensions of $C_{n,m}$ to the log case that allow boundaries on the base space, but we do not treat them in this note.

The purpose of this note is to prove Theorem 1.3. We also answer $\operatorname{Log} C_{2,1}$ affirmatively in positive characteristic.

Theorem 1.6. *Let k be an algebraically closed field of characteristic $p>0$. Let (S,Δ) be a projective lc pair of dimension 2 and let $f: S \to C$ be an algebraic fiber space to a smooth projective curve C. Then,*

$$\kappa(S, K_S+\Delta) \geq \kappa(S_\eta, K_{S_\eta}+\Delta|_{S_\eta}) + \kappa(C). \tag{I}$$

Here, S_η is the generic fiber of f.

Declaration of AI assistance. The author used OpenAI's ChatGPT during the development of the example presented in this paper. In particular, ChatGPT was used to explore possible constructions and to assist with preliminary mathematical calculations. All mathematical statements and proofs presented in this paper have been independently verified by the author, who takes full responsibility for their correctness.

2. Counter-examples to $\operatorname{Log} C_{3,2}$

In this section, we construct counter-examples to $\operatorname{Log} C_{3,2}$. Let k be an algebraically closed field of characteristic $p=2$ or 3. We use the construction of a Tango–Raynaud curve in [18, Example 1.3].

Put $l:=5-p$. Let e, $C=Y$ and D be as in [18, EXAMPLE 1.3] with $l|e$. Take $\xi \in H^1(Y, \mathcal{O}_Y(-D)) \setminus \{0\}$ such that $F^*\xi = 0$. Let

$$0 \to \mathcal{O}_Y(-D) \to \mathcal{E} \to \mathcal{O}_Y \to 0$$

be the extension corresponding to ξ. Then, by $F^*\xi=0$, we find a subbundle $\mathcal{O}_Y \subset F^*\mathcal{E}$. Set $P:=\mathbb{P}(\mathcal{E})$ and let $g: P \to Y$ be the projection. Put $T:=c_1(\mathcal{O}_P(1))$. Let F be the prime divisor on P corresponding to the subbundle $\mathcal{O}_Y(-D) \subset \mathcal{E}$. Let G' be the prime divisor on $\mathbb{P}(F^*\mathcal{E})$ corresponding to the subbundle $\mathcal{O}_Y \subset F^*\mathcal{E}$. Set $G:=F^*_{P/Y}G'$, where $F_{P/Y}$ is the relative Frobenius morphism of g. Then, by [18,

PROPOSITION 1.7], we see that $\mathrm{Supp}(G)$ is smooth. Since $F \sim T + g^*D$ and $G \sim pT$, we have

$$F + G \sim (p+1)T + g^*D = l(rT + g^*D'),$$

so we can take an l-cyclic cover $\pi : S \to P$ branched along $F + G$, where $r := \frac{p+1}{l}$ and $D' = \frac{1}{l}D$ is an integral divisor. This S is called a Raynaud surface. Note that, if Y is a supersingular elliptic curve and $D = 0$, then S is a quasi-hyperelliptic surface. Also, we note that

$$K_P + \frac{2}{p}G \sim_{\mathbb{Q}} -2T + g^*(K_Y - D) + 2T \sim g^*(K_Y - D) \quad \text{and}$$
$$K_S \sim \pi^*(K_P + (l-1)(rT + g^*D')) \sim h^*(K_Y - D') = h^*((pl-1)D'),$$

where $h : S \to Y$ is the natural morphism. Therefore,

$$\kappa(S) = \begin{cases} 0 & \text{if } D' = D = 0 \\ 1 & \text{otherwise.} \end{cases}$$

Note that $K_Y = pD = plD'$ by the construction of Y.

Put $X' := P \times_Y S$ and $B' := G \times_Y S \subset X'$. Then X' is smooth and B' is not smooth over the non-smooth locus of $h : S \to Y$. One can easily check that

$$K_{X'} + \frac{2}{p}B' \sim_{\mathbb{Q}} \sigma^*(K_Y - D - D'),$$

where $\sigma : X' \to Y$ is the natural morphism. Set $Z := \mathrm{Sing}(B')_{\mathrm{red}} \subset X'$. Let $\mu : X \to X'$ be the blow-up along Z, let B be the strict transform of B' and let E be the exceptional prime divisor of μ. We prove the following:

(1) X and B are smooth, and
(2) putting $c := \frac{p-4}{p} < 0$, we get

$$K_X + \frac{2}{p}B = \mu^*\left(K_{X'} + \frac{2}{p}B'\right) + cE.$$

If (2) holds, then we see that $\kappa\left(X, K_X + \frac{2}{p}B\right) < 0$, since $K_{X'} + \frac{2}{p}B'$ is $\mathbb{Q}$-linearly equivalent to the pull-back of a divisor on Y, $\mathrm{Supp}(E)$ dominates Y and $c < 0$. Since Z does not dominate S, we can easily check that $\kappa(X_\eta, K_{X_\eta} + \frac{2}{p}B|_{X_\eta}) = 0$.

Let us show (1) and (2). The local equation of S over G can be written as

$$u^l = x^p - t,$$

where t is a local parameter on Y. Let z be a local fiber coordinate on P, and put $w = z - x$. Then the equation of pr_1^*G is

$$u^l + w^p = 0.$$

By an easy calculation, we see that B is smooth. Note that X' and Z are smooth by the construction, so X is smooth. Since B' has multiplicity 2 along Z, and the codimension of $Z \subset X'$ is 2, we obtain

$$c = 2 - 1 - 2 \cdot \frac{2}{p} = \frac{p-4}{p}.$$

This completes the proof.

3. Proof of $\operatorname{Log} C_{2,1}$

In this section, we prove Theorem 1.6.

Proof. For simplicity, we denote $K_S + \Delta$ by K.

(1) We may assume that $\kappa(S_\eta, K|_{S_\eta}) \geq 0$ and $\kappa(C) \geq 0$.

(2) Suppose that $\kappa(S_\eta, K|_{S_\eta}) = 1$. Then $K|_{S_\eta}$ is ample. Taking a log resolution, we may assume that S is smooth. Put $\Delta_e := \frac{p^e}{p^e+1}\Delta$ for $e \gg 0$. Then
- $\Delta_e \leq \Delta$,
- $K_{S_\eta} + \Delta_e|_{S_\eta}$ is ample and
- $K_S + \Delta_e$ is a $\mathbb{Z}_{(p)}$-divisor.

Since $(S_\eta, \Delta|_{S_\eta})$ is F-pure, applying [7, Theorem 7.3], we get

$$\kappa(S, K) \geq \kappa(S, K_S + \Delta_e) \geq \kappa(S_\eta, K_{S_\eta} + \Delta|_{S_\eta}) + \kappa(C) = \kappa(S_\eta, K|_{S_\eta}) + \kappa(C).$$

(3) Let us consider the case when $\kappa(S_\eta, K|_{S_\eta}) = 0$. We use Tanaka's theorems on MMP for surfaces [20]. Firstly, taking a log resolution of (S, Δ) before running the MMP, we may assume that S is $\mathbb{Q}$-factorial. Write $\Delta = \Delta_h + \Delta_v$, where Δ_h and Δ_v are the horizontal part of Δ and the vertical part of Δ, respectively. Since $\kappa(S, K) \geq \kappa(S, K_S + \Delta_h)$, we may assume that $\Delta_v = 0$. Replacing S by its minimal model over C and applying the abundance theorem, we may assume that K is semi-ample over C. Since $\kappa(S_\eta, K|_{S_\eta}) = 0$, we see that $K \sim_{\mathbb{Q}} f^*L$ for some $\mathbb{Q}$-divisor L on C. To show inequality (I), it is enough to show that

$$\kappa(L) \geq \kappa(C).$$

(4) We consider the case when $\Delta_h = 0$. Let $\tilde{S}$ be a log resolution of S. Then applying [4, Theorem 1.3], we get

$$\kappa(S, K) \geq \kappa(S, K_S) \geq \kappa(\tilde{S}) \geq \kappa(\tilde{S}_\eta, K_{\tilde{S}_\eta}) + \kappa(C) = \kappa(S_\eta, K|_{S_\eta}) + \kappa(C).$$

(5) Next, we consider the case when $\Delta_h > 0$. We assume that $(S_\eta, \Delta|_{S_\eta})$ is klt. Let A be an effective ample divisor and choose an $\varepsilon \in \mathbb{Q}$ with $0 < \varepsilon \ll 1$. Then $(S_\eta, (\Delta + \varepsilon A)|_{S_\eta})$ is F-pure and $K_{S_\eta} + (\Delta + \varepsilon A)|_{S_\eta}$ is ample, so we see from (2) that $\kappa(S, K + \varepsilon A) \geq 0$. Taking $\varepsilon \to 0$, we see that K is pseudo-effective. Therefore, we obtain that $\deg(L) \geq 0$.

(6) We treat the case where $(S_\eta, \Delta|_{S_\eta})$ is not klt but lc. Let D be a curve on S such that $\operatorname{coeff}_D(\Delta) = 1$. Since D dominates C, we have

$$0 \leq \deg K_C \leq \deg K_D = (K_S + D) \cdot D \leq K \cdot D = \deg((f^*L)|_D),$$

so $\deg(L) \geq 0$.

(7) If $\deg(L) > 0$, then $\kappa(L) = 1 \geq \kappa(C)$, which completes the proof. If $\deg(L) = 0$, then by the abundance theorem, we obtain $\kappa(L) = 0$. Therefore, our last goal is to show that $\kappa(C) = 0$ under the assumption that $K = K_S + \Delta \sim_{\mathbb{Q}} 0$. When $(S_\eta, \Delta|_{S_\eta})$ is not klt but lc, we see from the above inequality that

$\deg K_C = 0$, so $\kappa(C) = 0$. Thus, we may assume that $(S_\eta, \Delta|_{S_\eta})$ is klt. Let D be a curve with $\delta := \mathrm{coeff}_D(\Delta_h) > 0$. Then

$$0 = K \cdot D = ((1-\delta)K_S + \delta(K_S + D) + (\Delta - \delta D)) \cdot D \geq (1-\delta)K_S \cdot D,$$

so $K_S \cdot D \leq 0$. Since $\deg(K_D) \geq 0$, by adjunction, we get that $D^2 \geq 0$. From this, we see that Δ is nef (note that $\Delta = \Delta_h$). Thus,

$$-K_S = -K + \Delta \sim_{\mathbb{Q}} \Delta$$

is also nef. Then we obtain from [6, Theorem 4.1] that $-K_C$ is pseudo-effective, and hence $\kappa(C) = 0$. Although [6, Theorem 4.1] assumes that K_S is $\mathbb{Z}_{(p)}$-Cartier, the argument proving pseudo-effectivity only requires that K_{S_η} be $\mathbb{Z}_{(p)}$-Cartier. In our setting, K_{S_η} is Cartier. Also, note that [6, Theorem 4.1] requires that the geometric generic fiber $S_{\overline{\eta}}$ is F-pure. Since $\Delta_h > 0$, we see that $-K_{S_\eta}$ is ample, which means that $S_{\overline{\eta}} \cong \mathbb{P}^1$, so we can apply the theorem.

□

References

[1] C. Birkar, Y. Chen, and L. Zhang. Iitaka $C_{n,m}$ conjecture for 3-folds over finite fields. *Nagoya Math. J.*, 229:21–51, 2018.

[2] J. Cao and M. Păun. Kodaira dimension of algebraic fiber spaces over Abelian varieties. *Invent. Math.*, 207(1):345–387, 2017.

[3] P. Cascini, S. Ejiri, J. Kollár, and L. Zhang. Subadditivity of Kodaira dimension does not hold in positive characteristic. *Comment. Math. Helv.*, 96(3):465–481, 2021.

[4] Y. Chen and L. Zhang. The subadditivity of the Kodaira dimension for fibrations of relative dimension one in positive characteristics. *Math. Res. Lett.*, 22(3):675–696, 2015.

[5] S. Ejiri. Weak positivity theorem and Frobenius stable canonical rings of geometric generic fibers. *J. Algebraic Geom.*, 26:691–734, 2017.

[6] S. Ejiri. Positivity of anti-canonical divisors and F-purity of fibers. *Algebra Number Theory*, 13(9):2057–2080, 2019.

[7] S. Ejiri. Direct images of pluricanonical bundles and Frobenius stable canonical rings of generic fibers. *Algebr. Geom.*, 11:71–110, 2024.

[8] S. Ejiri and L. Zhang. Iitaka ' s $C_{n,m}$ conjecture for 3-folds in positive characteristic. *Math. Res. Lett.*, 25(3):783–802, 2018.

[9] O. Fujino. Algebraic fiber spaces whose general fibers are of maximal Albanese dimension. *Nagoya Math. J.*, 172:111–127, 2003.

[10] O. Fujino. On subadditivity of the logarithmic Kodaira dimension. *J. Math. Soc. Japan*, 69(4):1565–1581, 2017.

[11] O. Fujino. *Iitaka conjecture: an introduction*. Springer, 2020.

[12] K. Hashizume. Log Iitaka conjecture for abundant log canonical fibrations. *Proceedings of the Japan Academy*, 96(10):87–92, 2020.

[13] S. Iitaka. Genera and classification of algebraic varieties. 1. *Sûgaku*, 24:14–27, 1972.

[14] Y. Kawamata. Kodaira dimension of algebraic fiber spaces over curves. *Invent. Math.*, 66(1):57–71, 1982.

[15] Y. Kawamata. Minimal models and the Kodaira dimension of algebraic fiber spaces. *J. Reine Angew. Math.*, 363:1–46, 1985.

[16] J. Kollár. Subadditivity of the Kodaira dimension: fibers of general type. *Adv. Stud. in Pure Math.*, 10:361–398, 1987.

[17] S. Kovács and Z. Patakfalvi. Projectivity of the moduli space of stable log-varieties and subadditivity of log-Kodaira dimension. *J. Amer. Math. Soc.*, 30:959–1021, 2017.

[18] S. Mukai. Counterexamples to Kodaira ' s vanishing and Yau ' s inequality in positive characteristics. *Kyoto J. Math.*, 53(2):515–532, 2013.
[19] Z. Patakfalvi. On subadditivity of Kodaira dimension in positive characteristic over a general type base. *J. Algebraic Geom.*, 27:21–53, 2018.
[20] H. Tanaka. Minimal models and abundance for positive characteristic log surfaces. *Nagoya Math. J.*, 216:1–70, 2014.
[21] K. Ueno. *Classification Theory of Algebraic Varieties and Compact Complex Spaces*, volume 439 of *Lecture Notes in Mathematics*. Springer, Berlin, Heidelberg, first edition edition, 1975.
[22] E. Viehweg. Canonical divisors and the additivity of the Kodaira dimension for morphisms of relative dimension one. *Compos. Math.*, 35(2):197–223, 1977.
[23] E. Viehweg. Klassifikationstheorie algebraischer varietäten der dimension drei. *Compos. Math.*, 41(3):361–400, 1980.
[24] E. Viehweg. Die Additivität der Kodaira Dimension für projektive Faserräume über Varietäten des allgemeinen Typs. (German) [the additivity of the Kodaira dimension for projective fiber spaces over varieties of general type]. *J. Reine Angew. Math.*, 330:132–142, 1982.
[25] L. Zhang. Subadditivity of Kodaira dimensions for fibrations of three-folds in positive characteristics. *Adv. Math.*, 354:106741, 2019.

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